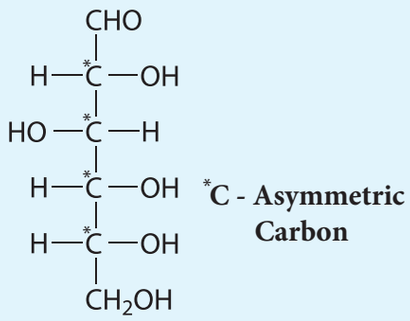
2. Glucose is produced commercially by the hydrolysis of starch with dilute HCl at high temperature under pressure.

$$\underset{\text{Starch}}{(C_6H_{10}O_5)_n} + n\,H_2O \xrightarrow[\substack{393K \\ 2\text{-}3\ atm}]{H^+} \underset{\text{Glucose}}{n\,C_6H_{12}O_6}$$

Structure of Glucose

Glucose is an aldohexose. It is optically active with four asymmetric carbons. Its solution is dextrorotatory and hence it is also called as dextrose. The proposed structure of glucose is shown in the figure 14.4 which was derived based on the following evidences.



D - (+)Glucose

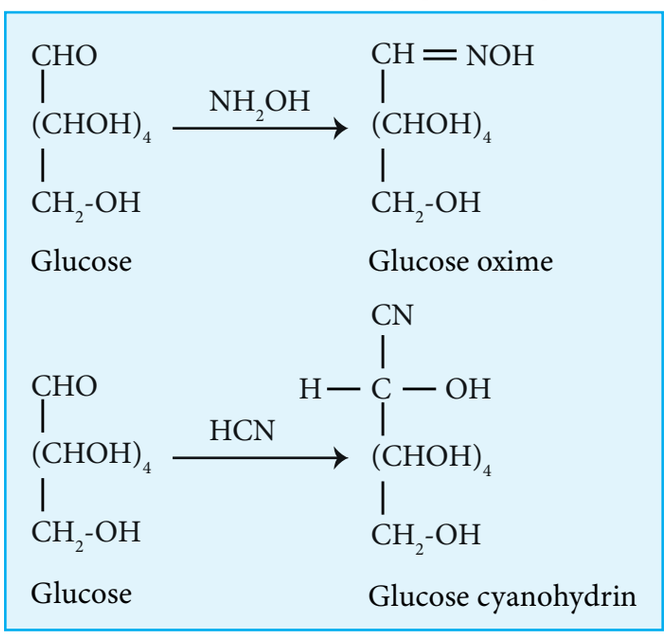
Figure 14.4 Structure of D(+) glucose

1. Elemental analysis and molecular weight determination show that the molecular formula of glucose is $C_6H_{12}O_6$.
2. On reduction with concentrated HI and red phosphorus at 373K, glucose gives a mixture of n hexane and 2–iodohexane indicating that the six carbon atoms are bonded linearly.

$$\text{Glucose} \xrightarrow[373\ K]{HI\ /\ P} CH_3\text{-}(CH_2)_4\text{-}CH_3 + CH_3\text{-}(CH_2)_3\text{-}\underset{\displaystyle I}{\underset{|}{CH}}\text{-}CH_3$$

n-hexane — **Major product**; 2-iodohexane — **Minor product**

3. Glucose reacts with hydroxylamine to form oxime and with HCN to form cyanohydrin. These reactions indicate the presence of carbonyl group in glucose.



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